PYTHON/STREAMLIT ESS SIMULATION

Validation Review and LSTM–Hybrid Comparison

Research: Development of AI-Enhanced Energy Storage System for Rural Nigeria

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1. Scope of this review

The review checks the forecast file against the established Version 3 results, aligns forecasts with the actual Kajola dataset and recorded solar-generation series, executes the core ESS dispatch calculations, and tests energy balances and battery operating limits. The forecast test period contains 2,632 hourly records from 13 September 2025 08:00 through 31 December 2025 23:00.

2. Corrected forecasting comparison

The previous prototype ZIP contained a forecast CSV whose values did not reproduce the approved Version 3 metrics. It has been replaced in the reviewed package with the Version 3 forecast file. The table below is recomputed from that corrected file.

Model	MAE (kW)	RMSE (kW)	MAPE (%)	R^2
Naïve persistence	2.1645	3.1403	15.0241	0.6963
ANN	4.3763	5.7163	30.4644	-0.0064
LSTM	1.0224	1.9228	7.3835	0.8861
Differencing–ANN–Bi-LSTM	1.2641	2.0632	9.5082	0.8689

Finding: LSTM remains the best model by the reported forecasting metrics. Its MAE is 1.0224 kW versus 1.2641 kW for the hybrid model. The hybrid must not be described as the most accurate model in this Version 3 test.

3. ESS simulation assumptions

The default comparison uses a 50 kWh battery, 15 kW maximum charging and discharging power, 50% initial state of charge (SOC), a 10%–95% SOC range, and 95% charge and discharge efficiencies. Recorded solar generation is aligned by timestamp. These are scenario assumptions, not verified specifications for a physical Kajola installation.

The reviewed controller serves actual load from solar first, charges from actual solar surplus, and uses the selected forecast to limit planned battery discharge. Actual observed demand is used to calculate realized unserved energy for both controllers. This is a transparent heuristic, not an optimized energy-management controller.

4. Results under default assumptions

Controller	Unserved with ESS (kWh)	Reduction vs no ESS (%)	Curtailed solar (kWh)	Losses (kWh)	Checks
LSTM	17,335.54	20.409	18,930.99	479.18	8/8
Hybrid (Differencing–ANN–Bi-LSTM)	17,335.84	20.407	18,931.32	479.14	8/8
Actual-load reference	17,330.08	20.434	18,924.95	479.77	8/8

The LSTM- and hybrid-guided results are very close under this particular rule. This is not evidence that their forecasting accuracy is equal; it indicates that the chosen dispatch heuristic and battery constraints produce similar outcomes for these two forecast series. The actual-load reference uses observed demand to guide discharge and is an idealized benchmark, not a deployable forecast controller.

5. Internal validation checks

For each controller, eight checks passed: load energy balance, solar energy balance, SOC transition balance, minimum SOC, maximum SOC, charging power limit, discharging power limit, and non-negative operating quantities. Maximum absolute load, solar, and SOC balance residuals were 0.000000 within numerical precision for the tested default scenarios.

These checks show internal accounting consistency under the implemented assumptions. They do not validate the model against physical battery measurements, and they do not establish real-world reliability improvements.

6. Capacity sensitivity

The prototype also compares 25, 50, 75 and 100 kWh capacities while holding the charge/discharge power limits at 15 kW and keeping the other assumptions fixed. Larger capacity generally reduced simulated unserved energy in the tested scenario. The results are scenario-dependent and should not be used to claim an economically optimal battery size.

Capacity	Controller	Unserved energy (kWh)	Unserved reduction (%)	Checks
25 kWh	LSTM	19,556.2	10.21	8/8
25 kWh	Hybrid	19,556.5	10.21	8/8
50 kWh	LSTM	17,335.5	20.41	8/8
50 kWh	Hybrid	17,335.8	20.41	8/8
75 kWh	LSTM	15,128.0	30.54	8/8
75 kWh	Hybrid	15,128.0	30.54	8/8
100 kWh	LSTM	12,961.8	40.49	8/8
100 kWh	Hybrid	12,961.8	40.49	8/8

7. Validation status and limitations

Completed: corrected Version 3 forecast file, timestamp alignment, recomputed forecasting metrics, core dispatch calculation checks, capacity sensitivity, and Python syntax compilation.

Not yet completed: a full interactive browser test of Streamlit in this environment, because Streamlit is not installed here; independent validation against measured battery/inverter data; validation of battery assumptions against a specific system; and supervisor approval of the proposed change from MATLAB/Simulink.

Conclusion: The reviewed Python core passes eight internal consistency checks for the default LSTM and hybrid scenarios. The current results support a controlled simulation comparison under stated assumptions only. They do not establish field performance or prove that either forecast improves real-world energy reliability.